

PAYSTREAM PAYMENT PROCESSING — PROCESS NOTES

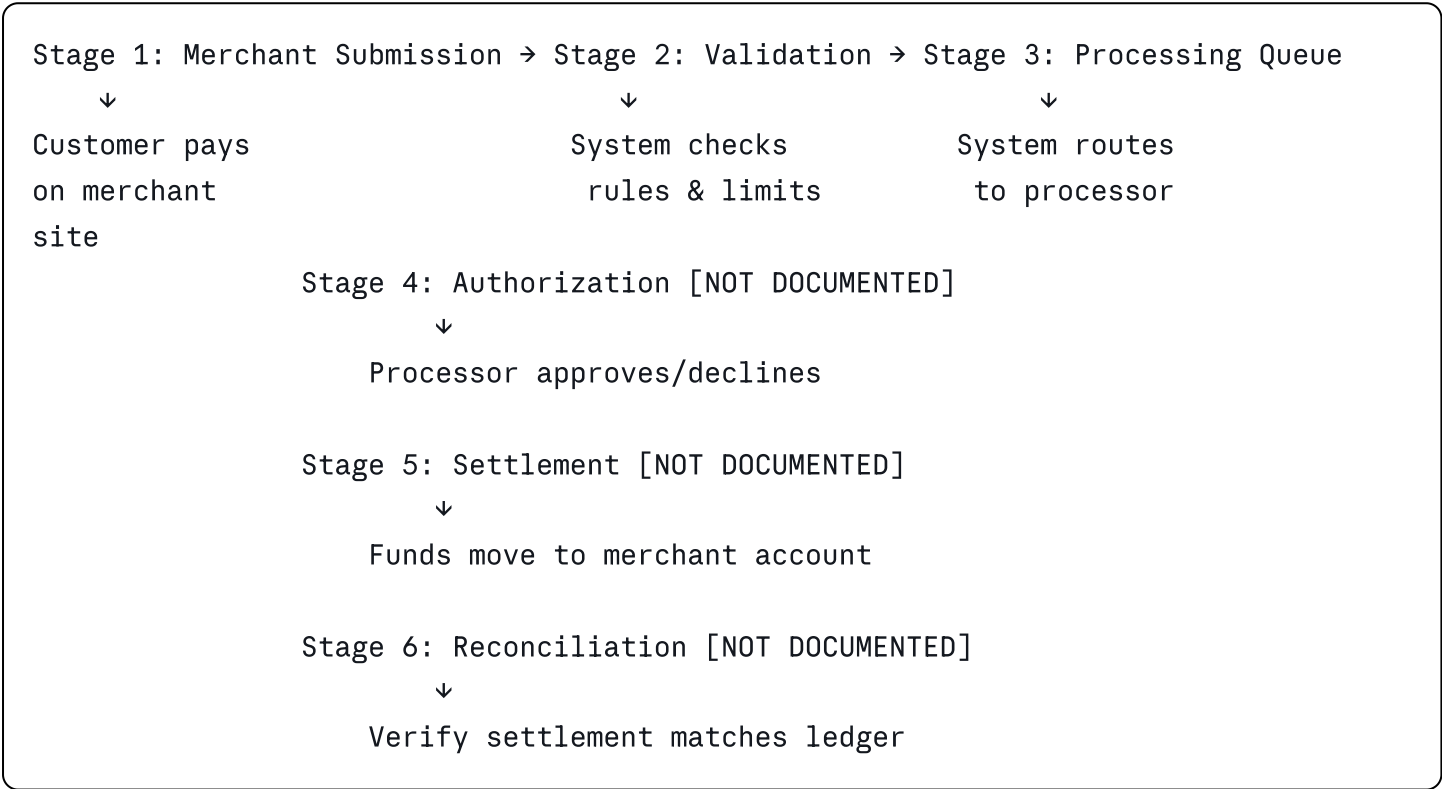
Last updated: 22 February 2026

Author: Tom Harrison (previous Data Analyst, now transferred)

STATUS: ⚠ INCOMPLETE — See "What's Not Documented" section below

PAYMENT PROCESSING FLOW — OVERVIEW

PayStream processes merchant payments in 6 stages (though only 1-3 are fully documented below).



STAGE 1: MERCHANT SUBMISSION

What happens: Customer enters payment details on merchant's checkout, submits transaction

Payment Methods (3 types):

1. **API_Direct** — Merchant integrates PayStream API directly into their system
 - Higher volume (60% of transactions)
 - Lower margin (1.1% fee) — used by larger SMEs and Mid-Market
 - Real-time feedback to merchant
 - Requires merchant to handle PCI compliance
2. **Merchant_Portal** — Customers use PayStream's hosted checkout page
 - Medium volume (25% of transactions)

- Higher margin (1.8% fee) — safer for smaller SMEs
- Customers redirected to PayStream, then back to merchant
- PayStream handles PCI compliance

3. **Bank_Transfer** — ACH/Faster Payments via bank

- Lower volume (15% of transactions)
- Variable fee (1.5% + fixed £0.30)
- Slowest settlement (3-5 days)
- Lowest failure rate

Data captured at Stage 1:

- merchant_id
 - payment_amount
 - payment_method
 - customer_email
 - billing_address (API_Direct only)
 - submission_timestamp
-

STAGE 2: VALIDATION

What happens: PayStream applies 4 validation checks before routing to processor

Validation Rule A: Merchant Account Status

Check: Is merchant account active and funded?

- If **NO** → Transaction fails (marked `failed_reason: MERCHANT_INACTIVE`)
- If **YES** → Continue to Rule B

Applied to: ALL payment methods

Failure rate historically: 0.8%

Validation Rule B: Amount Limits

Check: Is transaction amount within merchant's configured limits?

- Minimum: £0.99
- Maximum: Set per merchant (default £50,000)
- If **OUT OF RANGE** → Transaction fails (marked `failed_reason: AMOUNT_LIMIT`)

Applied to: ALL payment methods

Failure rate historically: 0.3%

Validation Rule C: Rate Limiting (Fraud Detection)

Check: Is customer attempting >5 transactions in 60 seconds from same IP?

- If **YES** → Block transaction (marked `failed_reason: RATE_LIMIT_EXCEEDED`)
- If **NO** → Continue

Applied to: API_Direct and Merchant_Portal only

Failure rate historically: 0.9%

Validation Rule D: Billing Validation Rule (BVR) ⚠

CHECK: Does merchant have sufficient funds AND valid billing info on file?

Rule Logic (CURRENT AS OF 22-FEB):

For API_Direct merchants:

- Check: `billing_address` is NOT NULL AND NOT BLANK
- Check: `account_balance` > `transaction_amount` × 1.5 (requires 150% reserve)
- Check: `no_failed_transactions_last_7_days` < 5

If ANY check fails → Block transaction

Marked: `failed_reason: BILLING_VALIDATION_FAILED`

Applied to: API_Direct payment method ONLY

Failure rate historically: 4.1% (as of 20-FEB)

⚠ **CAUTION:** This rule was **modified in Week 8** (17 February 2026) by engineering team. See "Engineering Changes" section below.

STAGE 3: PROCESSING QUEUE

What happens: Validated transactions enter a processing queue awaiting Stage 4 (authorization)

Queue Characteristics:

- Transactions processed in batches (hourly)
- Priority: Bank_Transfer < Merchant_Portal < API_Direct
- SLA: 95% of transactions must be processed within 2 hours of submission
- Queue monitored by: James Norton (Data Engineer) and ops team

Queue Health Metrics (from queue report):

- Average queue depth: 8,200 transactions (as of 21-FEB)
- Transactions stuck >4 hours: 12 (0.15%) — CHECK WITH JAMES
- Failed processing: 34 (0.41%) — reason unclear, escalated to engineering

Recent Queue Anomalies:

- 17 FEB 10:30 AM: Queue depth spike to 18,400 (engineering deployment that morning)
 - 18-22 FEB: Queue depth returned to normal levels, but **failure patterns changed** ← CHECK THIS
 - 23 FEB onwards: Data not yet available
-

STAGES 4-6: NOT DOCUMENTED ❌

Urgently needed: Process details for authorization, settlement, and reconciliation

Questions for James Norton:

1. What processor are we using (Stripe? Adyen? Custom)?
2. How do authorization failures get handled?
3. What's the settlement timeline (same-day, T+1, T+2)?
4. How do merchants access transaction history?

Questions for Sarah Ahmed:

1. Are there SLA targets for each stage?
 2. Who monitors Stage 4-6 daily?
 3. What's the escalation path if a processor timeout occurs?
-

ENGINEERING CHANGES — WEEK 8

Date: 17 February 2026, 10:30 AM UTC

Change: Billing Validation Rule (BVR) modifications

Engineer: Someone from Platform Team (need name)

Impact: UNKNOWN — needs investigation

What changed (from commit log snippets):

- Modified BVR threshold check
- Added "strict_billing_validation" flag
- Applied to API_Direct merchants only
- Rollout: Gradual (10% → 50% → 100% by 18 FEB 2:00 PM)

Why it was deployed:

- Email subject: "Reduce merchant chargeback risk"
- Ticket: CR-2987 (DEPRECATED?)
- Goal: Stricter validation to catch fraud earlier

⚠️ **CORRELATION:** BVR change on 17 FEB. Transaction failures spike on 18 FEB. **Coincidence?**

Action items:

- ☐ Get full commit details from engineering repo
- ☐ Get the change reason from whoever requested CR-2987
- ☐ Compare transaction failures before/after 17 FEB exactly
- ☐ Did the rule threshold get set too high?

GLOSSARY

Term	Definition
API_Direct	Merchant integrates PayStream API directly (high volume, low margin)
Merchant_Portal	PayStream hosts checkout (customer redirected)
Bank_Transfer	ACH/Faster Payments (slowest, lowest failure)
failed_reason	Tag marking why transaction failed (e.g., BILLING_VALIDATION_FAILED)
BVR	Billing Validation Rule — checks merchant funding and billing info
Queue depth	Number of transactions waiting to be processed
SLA	Service Level Agreement (e.g., "95% process within 2 hours")
Chargeback	Merchant disputes transaction with their bank, demands refund
CR-2987	Change Request ticket (see Engineering Changes)

CONTACT DIRECTORY

Name	Role	Contact	Knows About
Sarah Ahmed	Senior Ops Manager (YOUR MANAGER)	sarah.ahmed@paystream.co.uk	Overall SLAs, merchant relationships, Board reporting

Name	Role	Contact	Knows About
James Norton	Data Engineer	james.norton@paystream.co.uk	Data pipeline, queue metrics, Stage 3 operations, data quality
Priya Patel	Head of Compliance (FCA)	priya.patel@paystream.co.uk	Regulatory requirements, FCA reporting, audit trail
Marcus Lee	Chief Product Officer	marcus.lee@paystream.co.uk	Merchant experience, product roadmap, customer communication
Tom Harrison	Previous Data Analyst (TRANSFERRED)	tom@[external email]	Created these notes, may have context on Stages 4-6
Platform Team	Engineering (BVR change)	engineering@paystream.co.uk	Billing Validation Rule, CR-2987, change reasons

WHAT'S NOT DOCUMENTED — OPEN QUESTIONS

Critical Gaps

- ☐ **Stage 4 Authorization:** How do we communicate with the processor? What's the API format?
- ☐ **Stage 5 Settlement:** When do merchant accounts get credited? What's the fee calculation formula?
- ☐ **Stage 6 Reconciliation:** How do we verify settlement? What's the audit trail?
- ☐ **Failure Handling:** If a transaction fails in Stage 4-6, what's the retry logic? Does merchant get notified immediately?

Data Questions

- ☐ Why are API_Direct failures suddenly high starting 18 FEB?
- ☐ Is the BVR rule threshold too strict? (150% reserve requirement)
- ☐ What's in the 12 transactions stuck in queue >4 hours?
- ☐ The 34 "failed processing" transactions in Stage 3 — what caused them?

Regulatory Questions (for Priya)

- ☐ Does the BVR rule change trigger FCA notification?
- ☐ What's our deadline for filing this in the transaction accuracy report?
- ☐ If we reverted the BVR rule, would we need to file an amendment?

Process Questions

- ☐ Is there a runbook for "transaction failure spike"?

- ☐ Who gets paged if failure rate exceeds 10%?
 - ☐ Are there other validation rules we should know about?
-

HOW TO USE THIS DOCUMENT

If you're the new analyst (YOU):

1. Read Stages 1-3 to understand the validated transactions you're analyzing
2. Identify which stage is failing (hint: look at the data patterns)
3. Use the open questions to guide your investigation
4. Reach out to James/Sarah/Priya with specific findings — they'll fill in Stages 4-6

If you need to brief stakeholders:

- Use the "Contact Directory" to know who owns what
 - Use the "What's Not Documented" section to show what you've discovered vs. what was unknown
 - Use the "Engineering Changes" section to show the timing correlation (17 FEB → 18 FEB spike)
-

DEPRECATED / OUTDATED

- **CR-1205:** Old fraud detection rule, replaced by Rate Limiting (Rule C) — DO NOT USE
 - **merchant_legacy_table.csv:** Old merchant database, **DO NOT REFERENCE** — use merchant_directory.csv instead
 - **Payment processor: "Worldpay Classic"** — We migrated to [CURRENT PROCESSOR — UNKNOWN, ASK ENGINEERING]
-

END OF NOTES

Status: This handover is incomplete. Stages 4-6 are critical but not yet documented.

Next steps:

1. Analyze Stages 1-3 data to find the failure pattern
2. Brief Sarah with findings from Phase 2 (Wednesday)
3. Based on those findings, James + engineering can help you understand Stages 4-6

Good luck.

— Tom Harrison

Data Analyst (transferred)

22 February 2026